

Passenger_id	Details						Status				
	baggage_id	flight_number	airline	departure_airport	destination_airport	check_in_time	status	weight	status	dimensions	additional_details
1001	77987987	EK77	XYZ AIRLINE	JKF	MAA	XX:XX:XX	DXB				

Figure 7: Baggage tracking information

tag ID, airport codes, time stamps, and status updates, is stored in EMR HBase. Although a lot of tables are created for BMS, we will focus on two key tables due to their enormous volume. Bagevent table will have 30 row entries per bag minimum and the 'baggage_tracking' table created and frequently updated using the HBase catalog. Partitioning and queries on the lakehouse can be made using Iceberg's HBase catalog.

EMR Apache Spark - real-time query: EMR Apache Spark processes the BIX data in real-time, leveraging the Spark catalog for seamless access to data sources. Spark jobs execute on the EMR cluster, enabling advanced analytics, real-time alerts, and visualisation of baggage tracking data.

The integration with the lakehouse architecture allows Spark to efficiently query and analyse data from both HBase and PostgreSQL.

The volume of airlines baggage data is really huge. The beauty of today's complex systems is that there is space for every technology. Adopting a lakehouse architecture that

combines modern cloud supported open source databases, Big Data and analytics features not only empowers the airline baggage management system to optimise baggage operations and improve the passenger experience, but also creates new opportunities for ground operations, efficient load planning, etc. **END** 

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branded experiences within virtual environments. Businesses can establish virtual storefronts or integrate their products and services into the fabric of the metaverse, engaging with users in novel and interactive ways. Virtual advertising spaces, sponsored events, and partnerships can drive revenue streams and establish brand presence in the metaverse.

Virtual services and consultation: Professionals can provide virtual consulting services, ranging from design and architecture to healthcare and education. Virtual experts can offer personalised guidance, training, and mentorship, leveraging the immersive and interactive nature of hyper-real platforms. The virtual service economy within the metaverse can encompass a wide range of industries, unlocking new revenue streams and expanding access to expertise on a global scale.

Platform development and infrastructure: The development and maintenance of hyper-real platforms

themselves present economic opportunities. Companies can specialise in building and maintaining the underlying infrastructure, including servers, networking, security, and content delivery platforms. Additionally, developers and creators can build and sell tools, plugins, and assets for hyper-real platforms, catering to the needs of

other businesses and users within the metaverse ecosystem.

As hyper-real platforms continue to evolve, the possibilities in the metaverse are boundless. From lifelike gaming adventures to immersive architectural walkthroughs and experiential learning, the future holds exciting transformations that will reshape our digital world **END** 

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